PROCEDURE

EMPIRICAL EVIDENCE

1 Case and stimulus *Section 3.1*

GeoDrama deployed in Nanjing's historic urban district: AI-generated, location-triggered narratives during a free walk without a predefined route.

Content	HC	PC
Expression	NC	NV
Interaction	PP	PIQ

Six perceived attributes as stimuli (S) of the SOR model

PLATFORM IN USE (SEE FIGURE 4)



Real-time positioning triggers the narrative of the nearby heritage site



Dispersed heritage sites recommended for continued exploration

2 On-site data collection *Section 3.2*

- **Recruited on site** · adults; verbal consent
- **Briefing** · operation and safety only; hypotheses not disclosed
- **GeoDrama walk** · approx. **10–15 min**
- **Questionnaire** · anonymous, on site, within **5 min**

Central Nanjing, 5 April–20 August 2026; platform and generation settings held constant

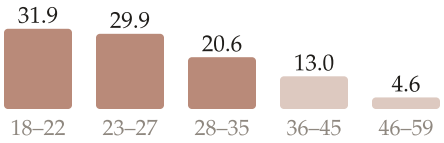
SAMPLE SCREENING AND PROFILE (TABLE 1)

607 participants completed the walk and questionnaire

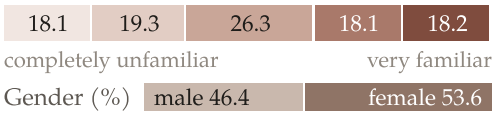
N = 548 valid — 90.3% retained — –59

Excluded: incomplete walk, system failure, straight-lining, low variance, contradictions

Age (%)



Familiarity with the area (%)



Gender (%) male 46.4 | female 53.6

3 Measurement and validation *Sections 3.3, 4.1*

27 items · 9 constructs (three items each; five-point Likert scale)

- Validated scales adapted; back-translation; wording reviewed by four experts
- Nine-factor CFA (AMOS 27.0): reliability, convergent and discriminant validity
- Harman's test and VIFs as preliminary checks

OBSERVED RANGE AGAINST CRITERION (TABLES 2–5)



CFA fit: $\chi^2/df = 1.166$ · CFI = .992 · TLI = .990 · RMSEA = .017 · SRMR = .029; VIFs 1.10–1.60

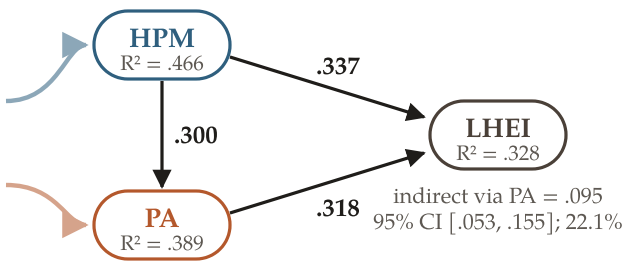
4 Structural model and alternative specifications *Sections 4.2–4.3*

- Maximum-likelihood SEM: direct paths H1–H14
- Three specifications differing only in the HPM–PA link (hypothesised, strict reverse, parallel), compared by AIC/BIC and χ^2 difference
- Bias-corrected bootstrap, **5,000** resamples: H15 and serial indirect associations

STANDARDISED ESTIMATES (TABLES 6–7, FIGURE 6)

	→ HPM cognitive	→ PA affective
HC	.285	n.s.
PC	.291	.224
NC	.171	n.s.
NV	n.s.	.157
PP	.153	.174
PIQ	.220	not hyp.

filled = $p \le .001$; size $\propto |\beta|$



HPM–PA link:

- HPM → PA: **AIC 509.8**, hypothesised, retained
- HPM ← PA: AIC 512.2, strict reverse, similar fit
- HPM · · · PA: $\Delta\chi^2 = 21.88$, $p < .001$, parallel, worse fit

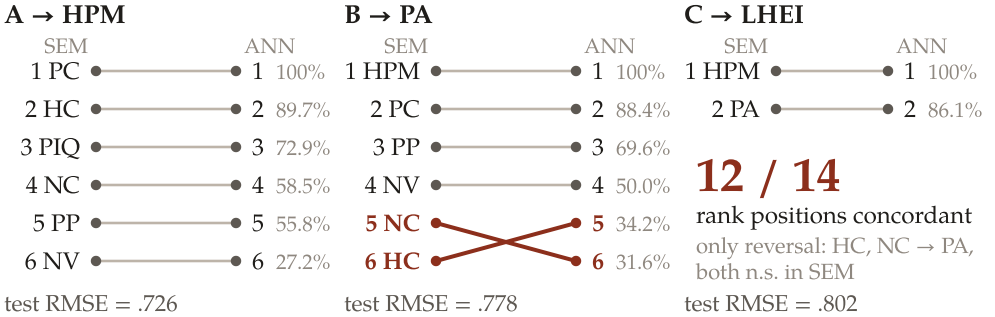
5 SEM–ANN predictive comparison *Section 4.4*

Three multilayer perceptrons (SPSS 27.0), one per SEM equation:

- **A** six attributes → HPM
- **B** HC, PC, NC, NV, PP and HPM → PA
- **C** HPM and PA → LHEI

Ten-fold cross-validation; predictors ordered by normalised importance and compared with the SEM β order

SEM RANK VS ANN RANK (TABLE 8)



12 / 14

rank positions concordant

only reversal: HC, NC → PA, both n.s. in SEM

% = mean normalised importance (ten folds); test RMSE ≤ training RMSE in all models

Synthesis *Sections 5–6*

Attribute–pathway matching

- HC, NC, PIQ → cognitive (HPM)
- NV → affective (PA) · PC, PP → both

Two organism states

HPM and PA structurally associated; each associated with LHEI

Implications

AI interpretation of dispersed, low-visibility heritage under SDG 11.4